

Optimizing User, Group, and Role Management with Access Control and Workflows

A PROJECT REPORT

Submitted by

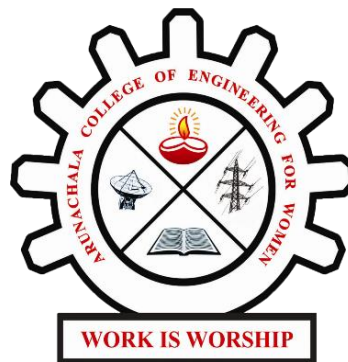
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In partial fulfillment for the award of the degree of

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ARUNACHALA COLLEGE OF ENGINEERING FOR WOMEN

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NOVEMBER 2025

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Abstract

The project titled “**Optimizing User, Group, and Role Management through Access Control and Workflows**” aims to design and implement a structured and secure access management system using the **ServiceNow** platform. In most organizational environments, unclear role assignments and poor access control mechanisms lead to inefficiencies, miscommunication, and data security issues. To overcome these challenges, this project focuses on establishing a systematic model that clearly defines users, groups, and roles, supported by automated workflows and Access Control Lists (ACLs).

The system is built around a simple project scenario involving a **Project Manager (Alice)** and a **Team Member (Bob)**. Each user is assigned specific roles and permissions to ensure that access to data and system functionalities aligns with their responsibilities. The **Project Manager** can create, assign, and approve tasks, whereas the **Team Member** can only view or update assigned tasks. By integrating **Users, Groups, Roles, Tables, ACLs, and Flow Designer**, the project establishes a seamless and transparent workflow that enhances accountability and collaboration.

During implementation, ServiceNow modules were configured to define users and groups, create custom roles, design Project and Task tables, assign permissions, and develop a workflow using Flow Designer. The workflow automatically updates task statuses, sends approval requests, and enforces access restrictions according to predefined rules. This automation significantly reduces manual intervention and potential human errors. Testing was carried out through user impersonation to verify that each role could perform only the intended operations.

The outcome of the project demonstrates improved security, efficiency, and clarity in access management. Users can now perform actions according to their designated roles, ensuring a controlled, transparent, and productive work environment. The project highlights how ServiceNow can be effectively utilized for **role-based access control (RBAC)** and **workflow automation**, ultimately optimizing team collaboration and governance. In the future, the system can be enhanced through integration with reporting dashboards, audit tracking, and automated alerts for policy violations to further strengthen operational intelligence and compliance.

1. Introduction

In organizations, poorly managed user permissions and unclear role assignments can lead to confusion, delays, and security risks. Without defined access boundaries, users may perform unauthorized actions or face difficulties collaborating effectively. To overcome these challenges, the “Optimizing User, Group, and Role Management” project was developed using **ServiceNow**, a powerful IT Service Management (ITSM) platform known for its flexibility and automation capabilities.

This project introduces a structured access control system by defining users (Project Manager and Team Member), assigning them to specific groups, and controlling their permissions through ACLs. Additionally, it incorporates an automated workflow to streamline task approvals and progress tracking. The primary objective is to build a secure, transparent, and efficient environment for user and role management.

In this project, the ServiceNow platform is utilized to demonstrate how automation and access control can simplify administrative processes. By leveraging ServiceNow’s modules - such as *User Administration*, *Groups*, *Roles*, *Access Control Lists (ACLs)*, and *Flow Designer* - the system ensures that only authorized users can access relevant data and perform actions based on their assigned responsibilities. This structured approach minimizes manual intervention, reduces administrative overhead, and enhances the overall transparency of operations within the organization.

Furthermore, the implementation of automated workflows provides real-time tracking and approval mechanisms, eliminating delays and communication gaps between teams. The system not only strengthens security but also improves collaboration and accountability by clearly defining user boundaries and approval hierarchies. Overall, this project showcases how an efficient combination of *access control and workflow automation* in ServiceNow can lead to a more secure, organized, and productive IT environment, aligning with the modern standards of enterprise service management.

2. Problem Statement

Before implementing this solution, the organization faced several issues due to manual handling of access and role assignments. There was no proper distinction between user permissions, resulting in data modification risks and miscommunication. Team members lacked visibility into their assigned responsibilities, and project managers found it difficult to track task completion efficiently. The absence of structured workflows caused delays and accountability gaps.

Therefore, the project aims to develop a **role-based access management system** in ServiceNow that standardizes user responsibilities, applies proper access control mechanisms, and automates the approval workflow - ensuring both security and productivity.

3. Methodology / System Design

i. Design Approach

The solution was developed using ServiceNow Studio, focusing on simplicity, clarity, and access security. The project defines users, groups, and roles, each with specific permissions. Access Control Lists (ACLs) are applied to restrict data modification based on user roles. The Flow Designer module is used to automate workflow approvals and task status updates, ensuring a seamless and efficient process.

ii. System Architecture

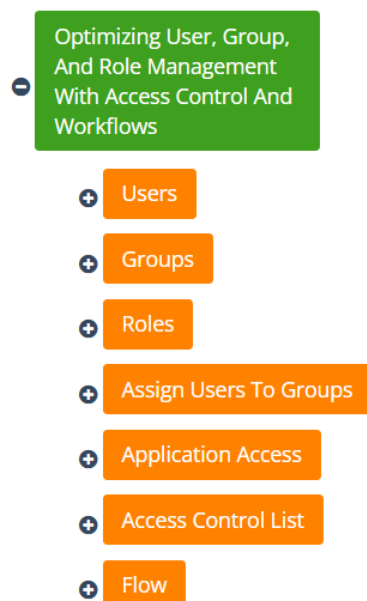
The architecture integrates the following ServiceNow components:

- **Users:** Project Manager (Alice) and Team Member (Bob)
- **Groups:** Project Team Group
- **Roles:** Project Manager Role, Team Member Role
- **Tables:** Project Table and Task Table
- **ACLs:** Role-based access permissions
- **Flow Designer:** Automated task approval and update workflow

Each component works together to form a secure structure. Alice (Project Manager) can create and assign tasks, while Bob (Team Member) can only view and update tasks assigned to him. Workflow automation ensures approvals and task status updates occur seamlessly.

iii. User Interface (UI) and User Experience (UX)

The interface is designed to be simple and consistent. The Project and Task tables are visible in the application menu, with form fields for project details, comments, and task status. Role-based visibility ensures that users only see data relevant to their permissions. A clean and structured layout enhances usability and workflow tracking.



4. Implementation Details

STEP 1: Create Users

1. Navigate to **All** → **Users** under *System Security*.
2. Create two users:
 - Alice (Project Manager)
 - Bob (Team Member)
3. Click **Submit** to save both users.

The screenshot shows the ServiceNow user management interface for a user named 'Alice p'. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The user's name 'User - Alice p' is displayed in the header. The form contains the following fields and options:

- User ID:** alice
- First name:** Alice
- Last name:** p
- Title:** (empty)
- Department:** (empty)
- Email:** alice@gmail.com
- Language:** -- None --
- Calendar integration:** Outlook
- Time zone:** System (America/Los_Angeles)
- Date format:** System (yyyy-MM-dd)
- Business phone:** (empty)
- Mobile phone:** (empty)
- Photo:** Click to add...
- Active:** ☒
- Password needs reset:** ☐
- Locked out:** ☐
- Web service access only:** ☐
- Internal Integration User:** ☐

Buttons at the bottom left: Update, Set Password, Delete. Below the form are links for 'Related Links' (View linked accounts, View Subscriptions, Reset a password) and a tabbed interface for 'Entitled Custom Tables' (Roles, Groups, Delegates, Subscriptions, User Client Certificates).

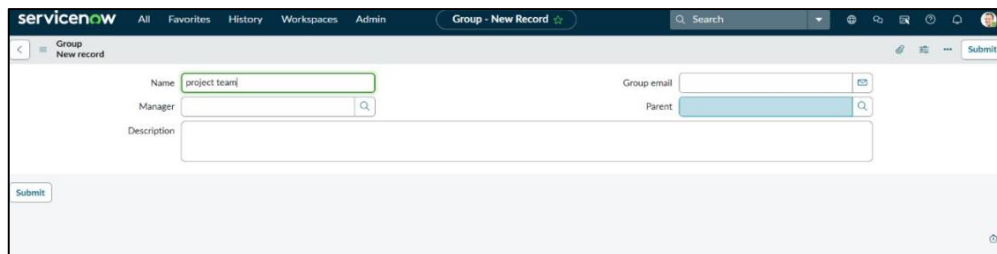
The screenshot shows the ServiceNow user management interface for a user named 'Bob p'. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The user's name 'User - Bob p' is displayed in the header. The form contains the following fields and options:

- User ID:** bob
- First name:** Bob
- Last name:** p
- Title:** (empty)
- Department:** (empty)
- Email:** bob@gmail.com
- Language:** -- None --
- Calendar integration:** Outlook
- Time zone:** System (America/Los_Angeles)
- Date format:** System (yyyy-MM-dd)
- Business phone:** (empty)
- Mobile phone:** (empty)
- Photo:** Click to add...
- Active:** ☒
- Password needs reset:** ☐
- Locked out:** ☐
- Web service access only:** ☐
- Internal Integration User:** ☐

Buttons at the bottom left: Update, Set Password, Delete. Below the form are links for 'Related Links' (View linked accounts, View Subscriptions, Reset a password) and a tabbed interface for 'Entitled Custom Tables' (Roles, Groups, Delegates, Subscriptions, User Client Certificates).

STEP 2: Create Groups

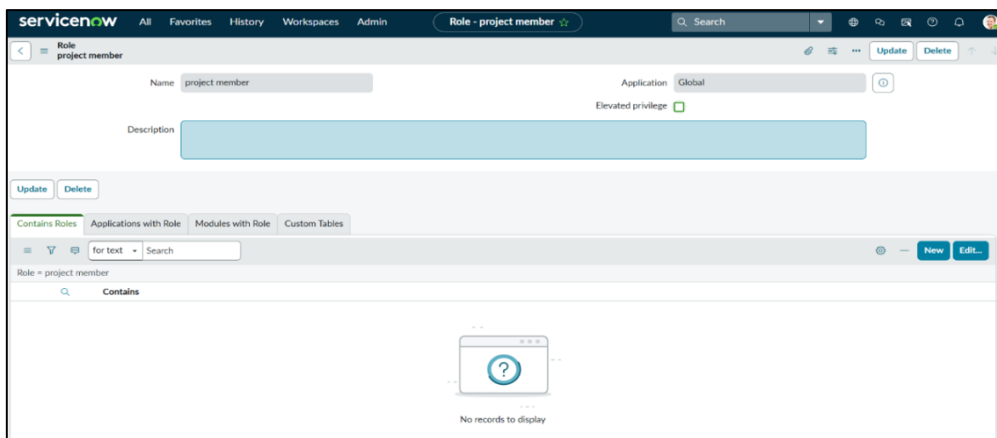
1. Go to **All** → **Groups**.
2. Create a new group named **Project Team**.
3. Add relevant details and click **Submit**.



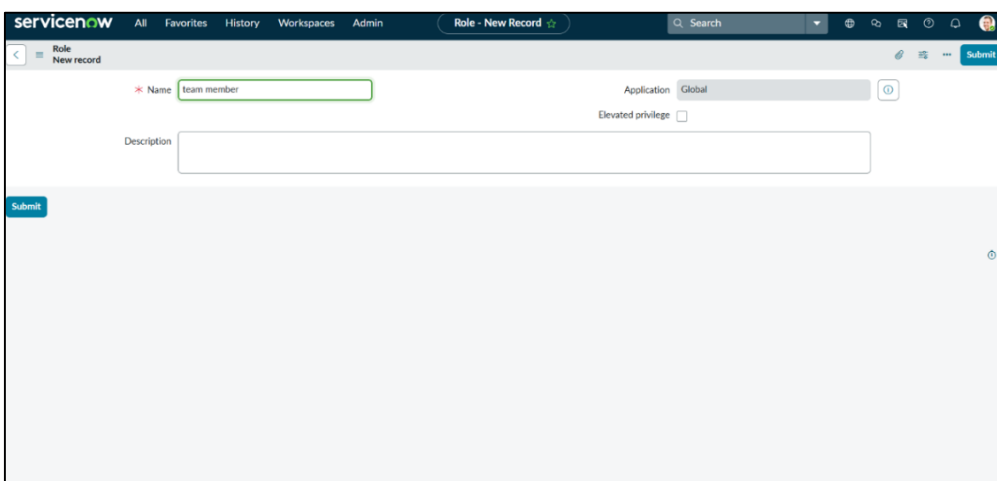
The screenshot shows the 'Group - New Record' form in ServiceNow. The 'Name' field is filled with 'project team'. The 'Manager' field has a search icon. The 'Group email' field is empty. The 'Parent' field has a dropdown menu. The 'Description' field is empty. A 'Submit' button is at the bottom left.

STEP 3: Create Roles

1. Navigate to **All** → **Roles**.
2. Create two roles:
 - **Project Manager Role**
 - **Team Member Role**
3. Click **Submit** after each role creation.



The screenshot shows the 'Role - project member' form in ServiceNow. The 'Name' field is filled with 'project member'. The 'Application' field is set to 'Global'. The 'Elevated privilege' checkbox is unchecked. The 'Description' field is empty. Below the form, there are tabs for 'Contains Roles', 'Applications with Role', 'Modules with Role', and 'Custom Tables'. The 'Contains' section shows 'No records to display'.



The screenshot shows the 'Role - New Record' form in ServiceNow. The 'Name' field is filled with 'team member'. The 'Application' field is set to 'Global'. The 'Elevated privilege' checkbox is unchecked. The 'Description' field is empty. A 'Submit' button is at the bottom left.

STEP 4: Create Tables

1. Go to **All** → **Tables** under *System Definition*.
2. Create a **Project Table** and enable *Create Module* and *Create Mobile Module*.
3. Add columns such as *Project Name*, *Assigned To*, *Status*, *Comments*, etc.
4. Similarly, create a **Task Table** with relevant fields.

The screenshot shows the 'Table - New Record' form in ServiceNow. The 'Name' field is set to 'u_project_table'. The 'Extends table' field is empty. The 'Create module' and 'Create mobile module' checkboxes are checked. The 'Add module to menu' dropdown is set to '-- Create new --'. The 'New menu name' field is set to 'project table'. The 'Remote Table' checkbox is unchecked. Below the form, the 'Columns' tab is active, showing a table of dictionary entries.

Column label	Type	Reference	Max length	Default value	Display
project id					false
project name					false
project manager					false

The screenshot shows the 'Table - New Record' form in ServiceNow. The 'Name' field is set to 'u_task_table_2'. The 'Extends table' field is empty. The 'Create module' and 'Create mobile module' checkboxes are checked. The 'Add module to menu' dropdown is set to '-- Create new --'. The 'New menu name' field is set to 'task table 2'. The 'Remote Table' checkbox is unchecked. Below the form, the 'Columns' tab is active, showing a table of dictionary entries.

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	{empty}	40	40	false
Updates	Integer	{empty}	40	40	false
Updated	Date/Time	{empty}	40	40	false
Sys ID		{empty}	32	40	false
		{empty}	40	40	false
		{empty}	40	40	false

servicenow All Favorites History Workspaces Admin Table - New Record

Table Columns Column label Search

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
X	Updated by	String	(empty)		40	false
X	Updates	Integer	(empty)		40	false
X	Updated	Date/Time	(empty)		40	false
X	Sys ID	Integer	(empty)		32	false
X	Created by	String	(empty)		40	false
X	Created	Date/Time	(empty)		40	false
X	task id	Integer				false
X	task name	String				false
X	assigned to	String				false
X	due date	Date				false
X	status	Choice				false
X	comments	String				false

Insert a new row...

Submit Cancel

Related Links

STEP 5: Assign Users to Groups

1. Open the **Project Team Group**.
2. Under *Group Members*, click **Edit**.
3. Add Alice and Bob to the group.
4. Save the changes.

servicenow All Favorites History Workspaces Admin User - alice p

User - alice p

Locked out ☐

Active ☒

Internal Integration User ☐

Business phone

Mobile phone

Photo Click to add...

Update Set Password Delete

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

Entitled Custom Tables Roles Groups (1) Delegates Subscriptions User Client Certificates

Role Search Edit...

User = alice p

Role	State	Inherited	Inheritance Count
No records to display			

STEP 6: Assign Roles to Users

1. Assign **Project Manager Role** and **Table Roles** to Alice.
2. Assign **Team Member Role** and limited **Task Table Role** to Bob.
3. Use the *Impersonate User* option to verify role-based access.

The screenshot shows the ServiceNow user profile for 'User - alice p'. The user is active, and the 'Internal Integration User' checkbox is unchecked. The 'Roles' tab is selected, showing a table of assigned roles.

Role	State	Inherited	Inheritance Count
u_project_table_user	Active	false	
u_task_table_2_user	Active	false	

The screenshot shows the ServiceNow user profile for 'User - bob p'. The user is active, and the 'Internal Integration User' checkbox is unchecked. The 'Groups' tab is selected, showing a table of assigned groups.

Group	State	Inherited	Inheritance Count
u_group_1	Active	false	

STEP 7: Application Access

1. Define which applications or tables each role can access.
2. Example: Project Manager → Full access to both tables; Team Member → Task Table only.

The screenshot shows the 'Application Menu - project table' configuration page in ServiceNow. The page has a dark header with the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, and Application Menu - project table. Below the header, there's a sub-header 'Application Menu project table' with 'Update' and 'Delete' buttons. A blue informational bar states: 'Restricts access to the specified roles. Otherwise, all users can view the application menu when it is active.' The main form includes a 'Roles' field with a dropdown menu showing 'u_project_table_user'. Below this, a blue bar explains the 'menu category' field. The 'Category' field is set to 'Custom Applications'. There are 'Hint' and 'Description' text input fields. At the bottom, there are 'Update' and 'Delete' buttons. A table at the very bottom lists application menu items with columns: Title, Table, Active, Filter, Order, Link type, Device type, Roles, and Updated.

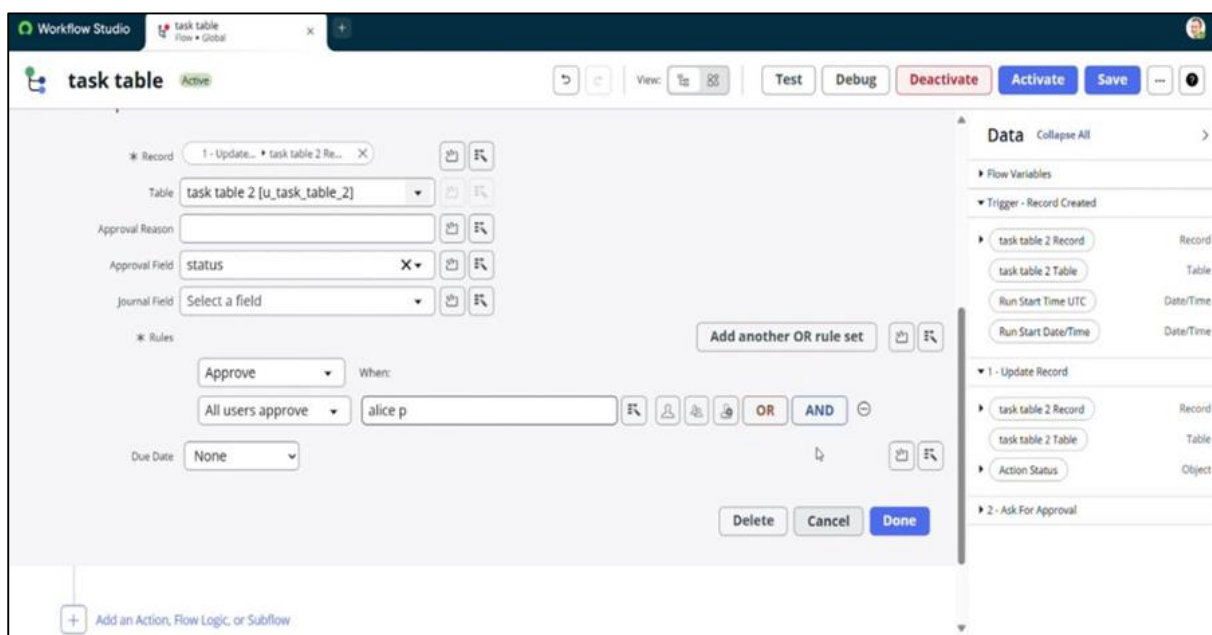
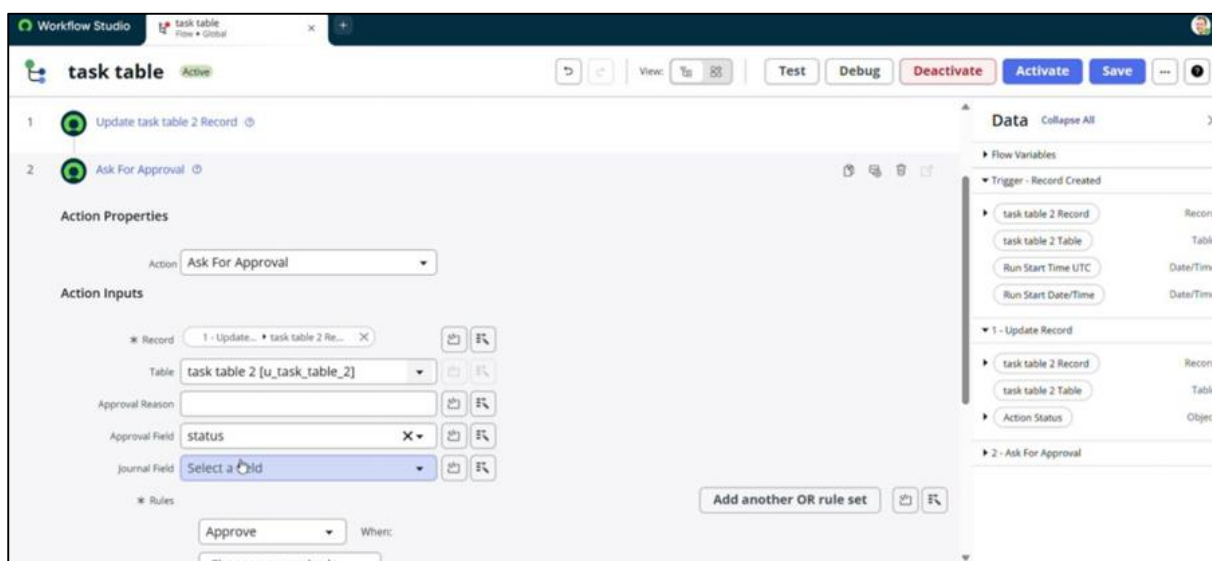
STEP 8: Access Control List (ACL)

1. Create ACL rules under *System Security* → *Access Control (ACL)*.
2. Restrict access to sensitive fields.
 - Example: Bob can edit *Comments* and *Status* only.
3. Test access by impersonating users to verify restrictions.

The screenshot shows the 'Access Control - New record' form in ServiceNow. The header includes the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, and Admin. The sub-header is 'Access Control New record' with a 'Submit' button. A blue warning bar at the top states: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' The form fields include: '* Type' (dropdown set to 'record'), 'Operation' (dropdown set to 'write'), 'Decision Type' (dropdown set to 'Allow If'), 'Application' (dropdown set to 'Global'), 'Active' (checkbox checked), and 'Advanced' (checkbox unchecked). There is an 'Admin overrides' checkbox checked. The 'Protection policy' is set to '-- None --'. The 'Name' field has a dropdown set to '*' and another dropdown set to '-- None --'. There is a 'Description' text input field. At the bottom, there are 'Applies To' buttons: 'Add Filter Condition' and 'Add OR Clause'. A 'Conditions' section is visible at the bottom of the form.

STEP 9: Workflow Implementation (Flow Designer)

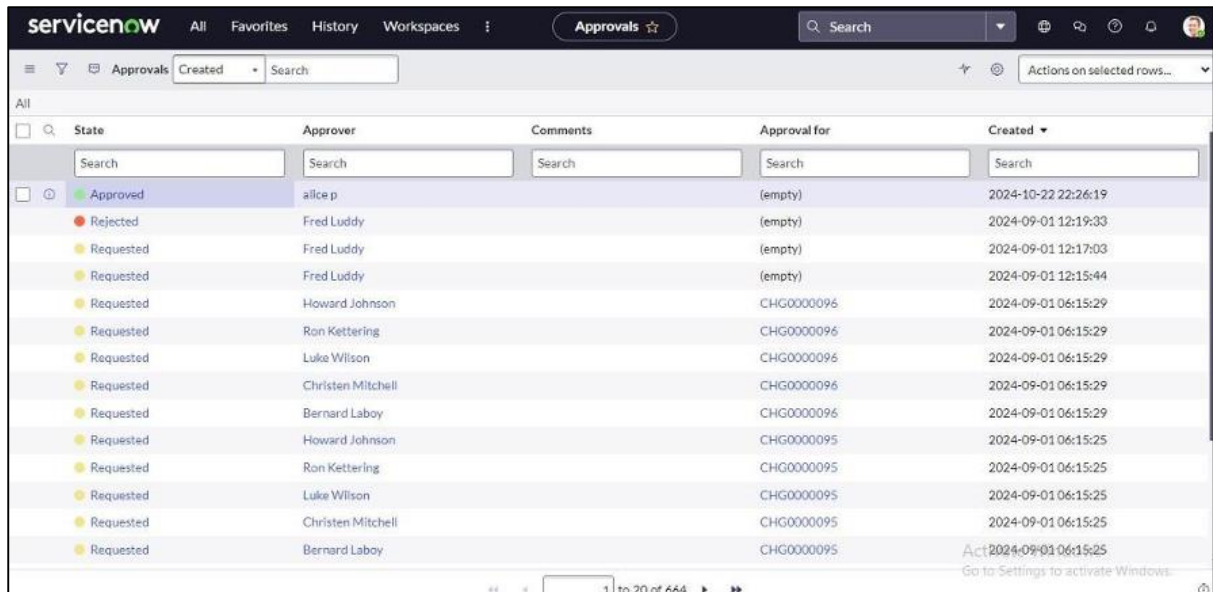
1. Open **Flow Designer** → **New Flow**.
2. Name the flow “Task Table Automation.”
3. Set Trigger: *Record Created or Updated* on the Task Table.
4. Add Conditions:
 - Status = “In Progress”
 - Assigned to = Bob
5. Add Actions:
 - **Update Record:** Change status to “Completed.”
 - **Ask for Approval:** Send approval to Alice.
6. Test the flow to ensure tasks are automatically updated and approved.



5. Testing

Testing was performed by impersonating both users:

- As **Bob (Team Member)**: Could view and update assigned tasks only.
- As **Alice (Project Manager)**: Could approve, monitor, and complete project tasks. The access control and workflow automation functioned correctly, ensuring secure and role-specific operations.



The screenshot displays the ServiceNow 'Approvals' page. The interface includes a top navigation bar with 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and an 'Approvals' button. A search bar is located on the right. Below the navigation bar, there's a filter section with 'Approvals' selected and a 'Created' filter dropdown. The main table lists approval tasks with columns: State, Approver, Comments, Approval for, and Created. The first row is highlighted in blue and shows an 'Approved' state with approver 'alice p' and a creation time of '2024-10-22 22:26:19'. Subsequent rows show 'Rejected' and 'Requested' states with various approvers like 'Fred Luddy', 'Howard Johnson', 'Ron Ketterling', 'Luke Wilson', 'Christen Mitchell', and 'Bernard Laboy'. The 'Approval for' column contains either '(empty)' or a reference ID like 'CHG0000096'. The 'Created' column shows timestamps from '2024-09-01 12:19:33' to '2024-09-01 06:15:25'. At the bottom, there's a pagination bar indicating '1 to 20 of 664' items.

	State	Approver	Comments	Approval for	Created
☐	Approved	alice p		(empty)	2024-10-22 22:26:19
☐	Rejected	Fred Luddy		(empty)	2024-09-01 12:19:33
☐	Requested	Fred Luddy		(empty)	2024-09-01 12:17:03
☐	Requested	Fred Luddy		(empty)	2024-09-01 12:15:44
☐	Requested	Howard Johnson		CHG0000096	2024-09-01 06:15:29
☐	Requested	Ron Ketterling		CHG0000096	2024-09-01 06:15:29
☐	Requested	Luke Wilson		CHG0000096	2024-09-01 06:15:29
☐	Requested	Christen Mitchell		CHG0000096	2024-09-01 06:15:29
☐	Requested	Bernard Laboy		CHG0000096	2024-09-01 06:15:29
☐	Requested	Howard Johnson		CHG0000095	2024-09-01 06:15:25
☐	Requested	Ron Ketterling		CHG0000095	2024-09-01 06:15:25
☐	Requested	Luke Wilson		CHG0000095	2024-09-01 06:15:25
☐	Requested	Christen Mitchell		CHG0000095	2024-09-01 06:15:25
☐	Requested	Bernard Laboy		CHG0000095	2024-09-01 06:15:25

6. Future Scope

The project “**Optimizing User, Group, and Role Management through Access Control and Workflows**” successfully implemented a structured role-based access control system and automated workflow management using the ServiceNow platform. While the present version efficiently manages users, groups, and roles with proper access control and task automation, there is significant scope for enhancement and scalability in future implementations.

One of the key areas for future development is the **integration of analytics and reporting modules** within the ServiceNow environment. This would allow administrators to track user performance, access patterns, and workflow efficiency through visual dashboards. Automated reports can help management analyze bottlenecks, monitor system usage, and make informed decisions to optimize team performance.

Another potential improvement lies in introducing **multi-level and dynamic approval workflows**. In larger organizations, projects often require approvals from multiple departments or hierarchies. By integrating adaptive workflows that automatically route requests based on user roles or departments, the system can handle complex approval structures more effectively and reduce manual oversight.

Additionally, **integration with external systems** such as email clients, inventory databases, or HR management tools can make the system more comprehensive. For instance, integrating with HR systems would allow automatic creation or deactivation of users based on employment status, ensuring consistent data management and security.

The implementation can also be extended to include **real-time notifications and AI-based access recommendations**. With machine learning models, ServiceNow can predict access requirements, detect anomalies in user activity, and suggest optimal role configurations, thus enhancing security and operational intelligence.

In the long term, the project can evolve into a **centralized access management solution** across various enterprise platforms, ensuring compliance, transparency, and efficiency at all levels. These future enhancements would make the system more robust, intelligent, and adaptable to the growing demands of digital enterprises, aligning with the vision of creating a smarter, automated, and secure workflow environment.

7. Conclusion

The project “**Optimizing User, Group, and Role Management through Access Control and Workflows**” successfully achieved its objective of establishing a systematic and secure access management system using the ServiceNow platform. Through the effective use of ServiceNow tools such as *Users, Groups, Roles, Tables, Access Control Lists (ACLs)*, and *Flow Designer*, the project streamlined administrative operations and improved control over user permissions within an organization.

The developed system ensures that every user receives the right level of access based on their assigned role and responsibilities. By automating approval workflows and integrating role-based access control mechanisms, the project minimizes manual errors, enhances transparency, and ensures accountability in task execution. This approach not only saves time but also strengthens data security by preventing unauthorized access and simplifying permission audits.

Furthermore, the project demonstrated how ServiceNow’s low-code/no-code capabilities can be leveraged to design efficient workflows that improve operational productivity. The testing phase confirmed that users, groups, and roles were managed accurately and that the automated workflows performed as expected, providing real-time updates and notifications at every stage.

Overall, this project serves as a practical example of how automation can transform administrative management into a more organized and reliable process. It highlights the importance of integrating technology-driven solutions in organizational access management and lays a strong foundation for future developments in workflow optimization and enterprise automation.

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